

Solving Inequalities with the Distributive Property

Teacher's Edition

Standard and Targets

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

Learning target I can solve inequalities that include the distributive property and represent the solutions on a number line.

Essential questions How do you solve a two-step inequality that includes parentheses? Where in a solution does the inequality symbol reverse, and why there?

Building on Lesson 8, and MA.7.AR.2.1.

Working toward MA.912.AR.2.6 — one-variable linear inequalities including compound inequalities.

Where this sits on the test Second of three consecutive lessons on MA.8.AR.2.2, a **scored Grade 8 FAST benchmark** in the Algebraic Reasoning category. Algebraic Reasoning scored lowest of the four reporting categories on the 2025 released items, and parentheses inside an inequality is the version of this benchmark most likely to appear as a two-part item.

Materials Whiteboards and markers. Calculators. Worksheet 7A.01.09. The two-method anchor chart from Lesson 5, which is still correct and now has one line added to it.

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: same and different	2	4 min
Notes I–IV: the same two methods, distributing a negative, dividing by a negative, checking a set	3–6	12 min
Example 1: which method, and does anything flip?	7–9	5 min
Example 2: the multiplier is negative	10–12	5 min
Example 3: the two sides the other way round	13–15	4 min
Whiteboards: 9 questions, each with its reveal	16–33	17 min
IXL — start today's skills	34	5 min
Total		53 min

Where the lesson lives: Notes II (slide 4) and Example 2 (slides 10–12). Notes II draws the line between rewriting one side and operating on both, which is the single idea that separates today from yesterday. Example 2 shows the same inequality flipping at two different moments and landing in the same place — the demonstration that stops the double flip.

Read this before you plan the day. Math Nation's components total 45 minutes — 5 for the Warm-Up, 10 for the Guided Instruction, 10 for the Guided Practice, 10 for the Collaboration, 5 for Your Turn! and 5 for the Wrap-Up — before any worksheet exists. The plan above fits periods because Sadie and Promus become Examples 1 and 2, the Collaboration's four-row error table becomes the whiteboard sequence and worksheet questions 1 and 2, and the Wrap-Up becomes the exit ticket.

No partner or group work anywhere in this lesson. Math Nation has three: the Collaboration is a partner error-analysis, the Guided Instruction tip asks partners to discuss before writing, and the Sadie/Promus comparison is framed as partner work. The error table is the best component in the lesson and it survives intact — it is simply worked by every student rather than split between two.

Vocabulary

distributive property — $a(b + c) = a \cdot b + a \cdot c$ — the factor outside multiplies every term inside

reverse — to turn the inequality symbol the other way round

solution set — every value that makes an inequality true

A note on the glossary. Math Nation lists no key terms and gives inequality and distributive property as additional terms. “Reverse” is added because the lesson's central question is where the symbol reverses, and students need one agreed word for it rather than four — flip, switch, turn, change.

Slide-by-Slide Teaching Notes

Slide 1. Inequalities with the Distributive Property 1 min

- Post the learning target.
- Nothing here is new — it is Lesson 5's two methods and Lesson 8's one rule, together for the first time.
- The difficulty is that a minus sign now appears in three different places and only one of them reverses anything.

Slide 2. Warm-Up 4 min

- TYPE: Same and different.
- INDEPENDENT.
- Expect: same — a -6 , a variable, both come from this unit.
- Different — the left is an expression to rewrite and the right is an inequality to solve; the left has no symbol to move and the right does.
- That distinction IS Notes II, and the warm-up is where they should meet it before it is named.
- OFF THE SLIDE, YOURS TO SAY: Both have a negative six in them. One is an expression to rewrite; the other is an inequality to solve.

Slide 3. Notes: I. The same two methods as Lesson 5 3 min

- Start from what they already have.

- The panel is the whole difference between this lesson and Lesson 5, and it is worth leaving on the board.
- OFF THE SLIDE, YOURS TO SAY: Same choice as Lesson 5. What is new is that one of the two roads has a flip on it.

Slide 4. Notes: II. Distributing a negative does NOT move the symbol *3 min*

- THE FIRST HALF OF THE NEW IDEA.
- Students who have just learned 'negative means flip' will flip on sight of the -6 .
- Say plainly what the reversal rule is actually about: an operation applied to both sides.
- Distributing is applied to one.
- OFF THE SLIDE, YOURS TO SAY: This is where students flip when nothing has happened.

Slide 5. Notes: III. Dividing by a negative DOES move it *3 min*

- THE SECOND HALF.
- The counting rule in item C is the practical test and it is what students should say out loud at every step this lesson.
- Two roads through the same problem can flip at different moments, or one can flip and the other not, and the answer is the same either way — which is the next slide.
- OFF THE SLIDE, YOURS TO SAY: Usually once. Sometimes not at all. Never twice for the same negative.

Slide 6. Notes: IV. Check a solution SET by testing two numbers *3 min*

- The check that catches a missed or a spurious reversal, and the reason it needs two values: substituting one number that happens to work proves nothing about the direction.
- Be precise about what the two outcomes mean.
- A reversed symbol makes the results SWAP: inside false, outside true.
- Two matching results have two possible causes and the benign one is not the one to reach for — yes, both values may have landed on the same side of the boundary, but two TRUES also mean the answer is missing solutions (too narrow) and two FALSES mean it contains numbers that are not solutions (too wide).
- Check the boundary first, and only re-pick once you are sure the two values really are on opposite sides of it.
- Do it on every example today.
- OFF THE SLIDE, YOURS TO SAY: One value cannot check an inequality. Two on the SAME side cannot either.

Slide 7. Example 1 *1 of the 5 min*

- SPOKEN ONLY.
- Method 2, because 3 divides 9 evenly; and nothing flips, because 3 is positive.
- Getting 'no flip' said out loud on a question where nothing flips is what makes 'flip' mean something two questions later.

Slide 8. Example 1 *2 of the 5 min*

- Work both columns.
- The panel is the reason to do it: two different routes, no reversal on either, same answer.

- That is the baseline the rest of the lesson departs from.
- OFF THE SLIDE, YOURS TO SAY: Positive 3 both times, so the symbol never moved on either road.

Slide 9. Example 1 *2 of the 5 min*

- Notes IV done once properly.
- Say what you would conclude if BOTH came out true: the value they called OUTSIDE is actually a solution, so their answer is too NARROW — it is missing solutions.
- That is the case an inside-only check cannot see, and it is why the outside value is asked for.
- A reversed symbol is the other pattern and item C already names it: inside false, outside true, the results SWAPPED.
- OFF THE SLIDE, YOURS TO SAY: One true, one false. That is what a correct solution set looks like.

Slide 10. Example 2 *1 of the 5 min*

- Take both answers.
- Method 2 flips at the very first step.
- Method 1 does not flip when distributing and flips later, at the division.
- Neither is wrong and the answers agree — that is the slide after next.

Slide 11. Example 2 *2 of the 5 min*

- THE SLIDE THIS LESSON IS FOR.
- Point at where each flip happened.
- Notes II is visible in Method 1's first line — the distribution rewrote a side and moved nothing.
- Notes III is visible in both last-but-one lines.
- OFF THE SLIDE, YOURS TO SAY: Same answer. One flip on each road, and it happens at a different line.

Slide 12. Example 2 *2 of the 5 min*

- The panel is worth pointing at.
- Example 1's answer was $x \geq 9$ and this one is $x \leq 3$, so 0 and 10 change sides.
- That is what a reversal looks like from the outside.
- OFF THE SLIDE, YOURS TO SAY: Compare with Example 1: the numbers that worked and failed have swapped over.

Slide 13. Example 3 *1 of the 4 min*

- Take a vote.
- Nothing flips — the 3 is positive and no operation on both sides involves a negative.
- The symbol only LOOKS reversed because the sides are written the other way round, which is Lesson 3's symmetric property.

Slide 14. Example 3 *2 of the 4 min*

- The last line is the one to be careful with.

- Turning $9 \geq x$ into $x \leq 9$ is the symmetric property and the symbol has to be re-pointed to keep the meaning — that is a rewrite, not a reversal, and a student who counts it as a flip will double-count on the worksheet.
- OFF THE SLIDE, YOURS TO SAY: The last line reads the same statement from the other end. That is not a flip.

Slide 15. Example 3 *1 of the 4 min*

- The summary slide, and it is the setup for the whiteboard round.
- Read the bottom row across and ask what is different about the middle column: the negative is attached to a number that both sides get divided by.
- That is the only case.
- OFF THE SLIDE, YOURS TO SAY: Three inequalities, one flip. Only a negative MULTIPLIER moves the symbol.

Slide 16. Whiteboards · Question 1 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 17. Whiteboards · Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline.
- No flip anywhere.
- Everything after this moves one minus sign.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 18. Whiteboards · Question 2 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 19. Whiteboards · Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the multiplier is negative.
- One flip, at the division.
- A board saying $x \geq 3$ did the arithmetic and skipped the rule.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 20. Whiteboards · Question 3 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 21. Whiteboards • Question 3 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT EARNS ITS PLACE.
- One feature changed: the minus is now on the VARIABLE, inside the bracket.
- Dividing by the positive 3 does not flip; dividing by the -1 in front of x at the very end does.
- Same one flip as question 2, arriving two lines later.
- A board that flipped at the first step has flipped for the wrong reason and will get the right answer for the wrong reason too — ask when, not just what.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 22. Whiteboards • Question 4 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 23. Whiteboards • Question 4 of 9 — reveal *about 1 minute with the discussion*

- One feature changed from question 1: the negative is on the RIGHT-HAND SIDE.
- Nothing is divided by a negative, so nothing flips.
- This is the one students flip out of habit — a minus sign appeared, and they reached for the rule without asking what it was attached to.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 24. Whiteboards • Question 5 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 25. Whiteboards • Question 5 of 9 — reveal *about 1 minute with the discussion*

- One feature changed from question 4: now BOTH are negative.
- Still only one flip, because only one division by a negative happened.
- Two minus signs do not mean two flips — Notes III's counting rule, and a board saying $x \geq 9$ flipped twice.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 26. Whiteboards • Question 6 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 27. Whiteboards • Question 6 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT CLOSES IT.

- One feature changed: the sides are swapped and every number is positive.
- SAME ANSWER as question 5, by a completely different route and with no flip at all.
- Put 5 and 6 side by side: one flipped once, one flipped never, and they agree.
- The symbol looking backwards is not the same as the symbol having moved.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 28. Whiteboards • Question 7 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 29. Whiteboards • Question 7 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: a constant now comes FIRST inside the bracket and the variable carries a negative coefficient.
- Dividing by the positive 3 changes nothing; dividing by the -2 at the end is the one flip.
- THE BOARD TO LOOK FOR SAYS $h > 1$, which is the arithmetic done and the rule skipped — and the substitution check catches it in one line, because $h = 2$ gives 9, which is not greater than 15.
- The other board to watch distributed first, got $21 - 6h > 15$, and is entirely correct; ask it how many flips it counted.
- Still one.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 30. Whiteboards • Question 8 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 31. Whiteboards • Question 8 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the sides are swapped AND every number is positive, which is question 6's shape with decimals in it.
- NOTHING FLIPS, and the answer still comes out as a $\geq$ because the letter ends up on the right and gets read back.
- THE BOARD TO LOOK FOR SAYS $x \leq -9$: that student saw the symbol apparently turn around when they rewrote $3 \leq 12 + x$ as $12 + x \geq 3$ and counted it as a flip.
- It is not a flip — it is the same statement read from the other end, which is question 6 of this round exactly.
- Reading an inequality backwards is free.
- Dividing by a negative is not.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 32. Whiteboards · Question 9 of 9 — ask *about 90 seconds of writing*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 33. Whiteboards · Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.
- With the worksheet gone this is the only place in the period the habit gets built, and IXL takes an answer without ever seeing the work.
- ANSWER: $y > 4$, with exactly ONE flip, at the first step, when both sides are multiplied by $-2/5$.
- THE MARK IS FOR THE ACCOUNTING, not the 4.
- This question has four minus signs and two fractions in it, and a student who writes " $y > 4$ " with no reasoning has either counted correctly or guessed, and nobody can tell which — including them.
- Every board that says $y < 4$ flipped a second time somewhere, and asking WHERE is what finds it: the $3/4$ is positive, the 6 is added, the 3 is divided by a positive.
- One negative, one flip.
- Substitution closes it: $y = 8$ gives -30 , which is below $-22\frac{1}{2}$; $y = 0$ gives -15 , which is not.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 34. IXL *5 min*

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."
- Makes them think." Circulate.
- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Worksheet 7A.01.09 — Answer Key

1. distribute first	$-30 \geq -48 + 6x$, $18 \geq 6x$, $x \leq 3$ NO reversal. The only division is by a positive 6. Distributing the -6 rewrites one side and leaves the comparison alone.
2. eliminate -6 first	$5 \leq 8 - x$, $-3 \leq -x$, $x \leq 3$ TWO reversals — once dividing by -6 , once multiplying by -1 . Two flips return the symbol to where it started, which is why both routes agree.
3. graph	<i>Closed circle at 3, shaded left.</i> The SOLVED line is $x \leq 3$, and its symbol has the equal bar, so the boundary belongs to the set.

4. verify	Inside $x = 0$: -48, and $-30 \geq -48$ ✓. Outside $x = 5$: -18, and $-30 \geq -18$ ✗. Boundary $x = 3$ gives exactly -30, and $-30 \geq -30$ ✓. An inside value on its own does catch a missed reversal — a reversed answer fails on its own inside value. What it cannot catch is an answer that is too NARROW, where everything inside really is a solution; that is what the outside value is for.
5. multiple choice	A $-(a + 4) > -7$ gives $-a > -3$, so $a < 3$ — open circle at 3, shaded left. B is what a missed final flip produces. C and D use a closed circle for a strict symbol.
6. $9 > 3(5c - 7)$	$c < 2$ open circle at 2, shade left Divide by 3 first: $3 > 5c - 7$. No flip anywhere.
7. $\frac{1}{3}(6 - m) \geq -4$	$m \leq 18$ closed circle at 18, shade left Multiply by 3 to get $6 - m \geq -12$. The flip arrives later, at $-m \geq -18$.
8. $-0.9(x + 2.5) < -2.25$	$x > 0$ open circle at 0, shade right Dividing by -0.9 reverses once, at the first step: $x + 2.5 > 2.5$. Distributing works too and reverses at the last step instead.
9. $(5/3)(11 - x/8) \geq 20$	$x \leq -8$ closed circle at -8, shade left $20 \div (5/3) = 12$, so $11 - x/8 \geq 12$ and $-x/8 \geq 1$. Distributing here is genuinely unpleasant; this item is the argument for Method 2.
10. two ways in words	Distribute: $-14 + 2d \leq -16$, so $2d \leq -2$, so $d \leq -1$. Divide by 2: $-7 + d \leq -8$, so $d \leq -1$. Neither way reverses, because 2 is positive. The two descriptions must be genuinely different methods, not the same steps renamed.

Questions 1 and 2 are the whole lesson in two questions. The same inequality, solved two ways: zero reversals one way and two the other, and both correct. Students who decide one route must be wrong because “the flips do not match” are reasoning from a rule they half-remember. The answer to give them is that flips are caused by divisions, not by the presence of a negative — count the divisions and the two routes stop looking contradictory.

Question 5's distractors are engineered. $-(a + 4) > -7$ becomes $-a > -3$, and the last step is a division by -1 . Choice B is what a student produces by skipping exactly that step. C and D check whether the strict symbol registered at all.

Question 10 is graded on the description, not the answer. Two genuinely different methods, in words, plus $d \leq -1$. A student who writes the same steps twice with different wording has not answered it. Neither method reverses anything here, and a student who writes that a flip occurs has told you something useful.

Additional Practice 7A.01.09 — Answer Key

Post this in Google Classroom alongside the practice sheet. It is optional work for students who want more.

1. distribute first	$21 - 6h > 15$, $-6h > -6$, $h < 1$ One reversal, dividing by -6.
---------------------	---

2. eliminate 3 first	$7 - 2h > 5$, $-2h > -2$, $h < 1$ Also one reversal, dividing by -2 . Dividing by the positive 3 at the start changes nothing.
3. graph	Open circle at 1, shaded left.
4. verify	Inside $h = 0$: $3(7) = 21 > 15$ ✓. Outside $h = 2$: $3(3) = 9 > 15$ ✗. Boundary $h = 1$ gives exactly 15, and $15 > 15$ ✗ — correct, since the symbol is strict.
5. multiple choice	B $-8 < \frac{2}{3}(c - 9)$ gives $-12 < c - 9$, so $c > -3$. Nothing reverses here — $\frac{3}{2}$ is positive — so a student who flips out of habit lands on C.
6. $6(g - 2) \leq 18$	$g \leq 5$ closed circle at 5, shade left
7. $21.6 \leq 7.2(12 + x)$	$x \geq -9$ closed circle at -9, shade right $21.6 \div 7.2 = 3$ exactly. Rewrite $-9 \leq x$ as $x \geq -9$ before graphing.
8. $-(\frac{5}{2})(\frac{3}{4}y + 6) < -22\frac{1}{2}$	$y > 4$ open circle at 4, shade right $(-22\frac{1}{2})(-\frac{2}{5}) = 9$, so $\frac{3}{4}y + 6 > 9$. The hardest computation on the sheet, and the reversal happens once, at the first step.
9. $(\frac{3}{8})(p + 29) > 9$	$p > -5$ open circle at -5, shade right $9 \div (\frac{3}{8}) = 24$.
10. two ways in words	Divide by -2.5 and reverse: $4 \leq z + 8$, so $z \geq -4$. Distribute: $-10 \geq -2.5z - 20$, so $10 \geq -2.5z$, so $z \geq -4$. Both ways reverse once and the -2.5 causes it both times. Distributing the -2.5 is not itself a reversal — that step only rewrites one side.

Misconceptions to Watch

- **Believing the factor beside the parentheses must always be distributed.** Named first in Math Nation's list. Notes I item C and the whiteboard sequence. (Math Nation TE)
- **When dividing first, dividing the factor into the terms inside the parentheses as well.** Also named — dividing by the number and then by the numbers inside. Example 1 Method 2. (Math Nation TE)
- **Forgetting to divide the other side when eliminating the factor.** Also named. (Math Nation TE)
- **Reversing the symbol when distributing a negative.** The double flip. Notes II, slides 10–12, worksheet questions 1 and 2.
- **Reversing at the end of every problem that contains a negative anywhere.** The whiteboard sequence, which includes a question with negatives on both sides and no multiplier among them.
- **Checking only a value inside the answer.** An inside-only check does catch a missed flip — a reversed answer fails on its own inside value. What it cannot catch is an answer that is too NARROW, where every value inside really is a solution and the missing ones are all outside. That is what the outside value is for. Notes IV and worksheet question 4.
- **Reading a stacked division step as a subtraction.** Math Nation prints “ $-7 \ -7$ ” beneath a line to mean divide by -7 . Write divisions as divisions.

Changes from the Math Nation Lesson

- A mislabeled number line was corrected. The Guided Instruction question 2 graph shows the solution to $x \geq 9$ with tick points reading $-8, 9, 10$. The first should be 8 .
- A model answer that named the wrong number was rewritten. Question 7 asks which method students prefer for $56 < 7(7 - x)$, and the printed sample answer says “I would rather get rid of the 3 in the beginning.” The 3 belongs to the previous problem. The same sample also argues both preferences in a single first-person voice.
- Division steps are written as divisions. Math Nation prints them stacked in the format of addition steps — a bare -7 under another -7 , or -1 under -1 — and students read them as subtracting. Every division step here says what it is.
- The verification question now asks for a value on each side. Math Nation’s substitution checks test only values inside the solution set, which cannot detect an answer that is too NARROW — every value inside a too-narrow answer satisfies the original, so an inside-only check passes it. Worksheet question 4 asks for one inside, one outside, and the key adds the boundary.
- A multiple-choice graph item was kept and its distractors rebuilt. Math Nation’s homework question 5 offers four unlabeled graphs; ours are labeled and each wrong choice corresponds to a specific error — a missed flip, a wrong circle, or both.
- Every partner and group activity is replaced — three of them. The Collaboration’s four-row error table is the strongest component in the lesson and it survives intact in the whiteboard sequence and worksheet questions 1 and 2, worked by every student rather than split between two.
- The word “homework” was removed. Math Nation’s tip reads “if time is an issue, then consider assigning 1.9.5 Your Turn! for homework.” Worksheet 7A.01.09 is classwork; the Additional Practice goes to Google Classroom for students who want more.
- Number lines are printed for every graphing item, scaled and labeled around the correct boundary.
- The Wrap-Up was replaced with a Ticket Out the Door that ends on the double flip — the one error this lesson exists to prevent.
- Answer keys were built for both sheets from the same source file as the student versions, so the two cannot drift apart.
- Nothing in the mathematics was changed.